



Make Your Own Smart Wireless Biometric Lock

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Traditional key-based locks are now outdated and come with limitations of their own. First, you always have to carry the key with you, and the key might get misplaced or stolen. Such locks are not completely secure either.

These old locks are now being replaced by modern biometric ones. However, biometric locks use biometric sensors that are expensive. Here is a cheap biometric lock using biometric fingerprint sensor available in your smartphone. Hence, no external biometric sensor is required. The prototype built is as inexpensive as possible and uses easily-available components. Hardware components used are listed in Table 1.

Arduino coding

First, write a program (biometric_lock.ino) in Arduino IDE that enables you

```
#include <Servo.h>
String reads;

Servo myservo;

void setup() {
  Serial.begin(9600);
  myservo.attach(9);
  // put your setup code here, to run once:
}
```

Fig. 1: Define servo in Arduino code

```
void loop() {
  while (Serial.available()==0);
  reads=Serial.readStringUntil('\n');

  if (reads=="adrfjhlmm")
  {
    myservo.write(0);
  }
  if (reads=="wrong")
  {
    myservo.write(79);
  }
}
```

Fig. 2: Arduino code checking device ID

Hardware Components

Table 1

Component name	Quantity	Description
Arduino Nano	1	For programming
Bluetooth HC 05	1	For wireless communication
Broken lock	1	You can get it from key maker
Servo motor	1	Micro servo motor
Wires	9	10cm
Battery	1	5V-6V

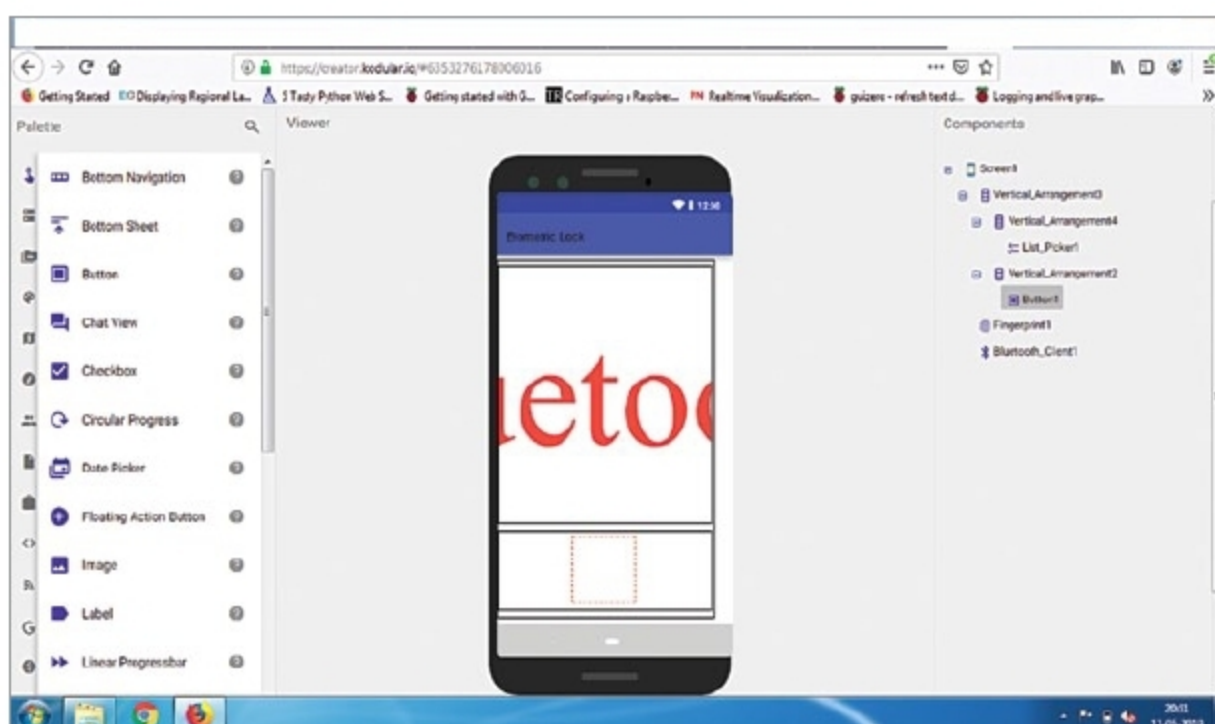


Fig. 3: App layout in Kodular

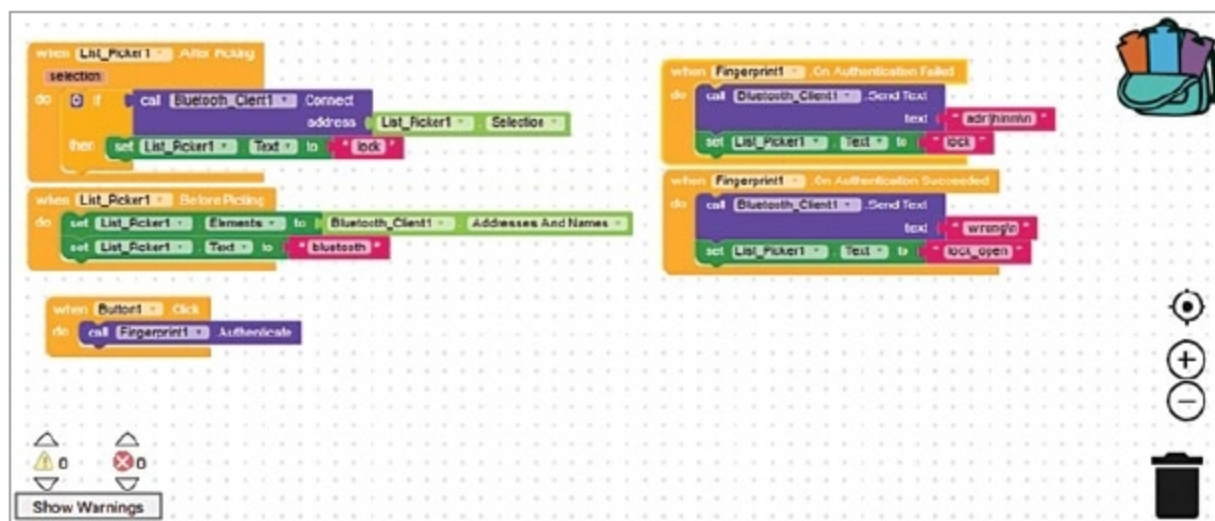


Fig. 4: Code blocks

to connect your smartphone with the lock and perform the task(s) according to the code. For that, create a string variable that stores the unique device ID for the lock; then, include the servo

library in your code.

Next, set serial baud rate to 9600 for HC-05 Bluetooth communication. You can change the baud rate if you are using a different Bluetooth